FlameLens – Innovation and Practical ML Strategy

Practical approach for handling limited NASA combustion datasets and building a scientifically defensible system.

1. The Main Problem with the Current Dataset

The current NASA data is useful for building a prototype, but it is not large or uniform enough to train strong general-purpose ML models for all three FlameLens modules.

The practical solution is not to artificially increase the number of rows or force every experiment into one model. Instead, each combustion problem should use the level of prediction that its available data can actually support.

Flame Extinction: suitable for the first real ML model because the FLEX dataset is directly related to droplet flame extinction.

Flame Spread: current BASS-II and SAFFIRE tables are not sufficient for a reliable general ML predictor, especially because the current tables do not provide a clean, consistent spread-rate target.

Soot/Smoke: SPICE has a useful number of experimental records, but a proper soot/smoke target needs to be identified from the analyzed experimental data before training a model.

The important principle is: do not create a prediction simply because the website has an input form for it. The model should only predict when the available experimental evidence supports the prediction.

2. Main Innovation – NASA Evidence + ML + Physics

The main innovation of FlameLens should be that it does not behave like a normal ML prediction website. It combines machine learning with NASA experimental evidence, data-coverage checking and physics-based interpretation.

Instead of only showing a prediction such as 'Extinction Probability = 72%', the system should also tell the user whether the requested condition is well represented by the available NASA experiments.

The proposed flow is:

User enters experimental conditions.

FlameLens checks whether similar NASA experiments exist.

If the condition is sufficiently represented, the ML model generates a prediction.

The system finds the closest NASA experiments used as evidence.

The model identifies important input factors.

The physics/RAG layer explains the result using the NASA evidence and relevant scientific information.

If the requested condition is outside the reliable data region, the system warns the user instead of giving a misleading prediction.

3. Prediction Availability / Data Coverage

A major feature can be a Prediction Availability or Data Coverage score. This allows FlameLens to know when it has enough evidence to make a prediction.

For example:

Prediction: High extinction risk
Model probability: 72%
NASA evidence: 8 similar experiments
Data coverage: High
Prediction status: Supported by existing NASA data

For an unsupported condition:

Prediction unavailable or low confidence.
Reason: The requested condition is outside the region well represented by the NASA experiments.
Nearest NASA experiments: FLEX #43, FLEX #51, FLEX #82.
Data coverage: Low

This is scientifically safer and makes the limited dataset itself part of the system's functionality.

4. Similar NASA Experiment Search

For every prediction, FlameLens should show the most similar NASA experiments. A normalized distance measure or K-Nearest Neighbors approach can be used to find experiments with similar experimental conditions.

The user sees which NASA experiments are closest to the requested condition.

The system can show the experimental conditions and observed outcome of those experiments.

The similarity result also gives a simple explanation for why the ML prediction is considered reliable or uncertain.

This feature can work even when a particular module does not yet have enough data for ML.

5. What-If Experiment Analysis

Instead of giving only one prediction, FlameLens can allow the user to change one experimental condition and observe how the predicted behaviour changes.

Example:

Oxygen concentration: 18% → predicted extinction probability 72%
Oxygen concentration: 20% → predicted extinction probability 51%
Oxygen concentration: 22% → predicted extinction probability 34%

This turns FlameLens into an exploratory research tool rather than a simple classifier.

6. Physics-Aware Explanation

The explanation should not be generated by an LLM without evidence. The ML model should first provide the important features, and the system should retrieve relevant NASA experiments and scientific sources.

Recommended architecture:

ML = prediction

SHAP or feature importance = important factors behind the prediction

NASA experimental records = evidence

RAG/LLM = readable scientific explanation

For example, the final explanation can say that the prediction was mainly influenced by oxygen concentration, pressure and initial droplet diameter, while also showing the NASA experiments that are most similar.

7. Data Gap Detection

FlameLens can visualize where NASA has a lot of experimental data and where the dataset is sparse.

For example, the system can show the distribution of experiments across oxygen concentration, pressure, airflow and droplet diameter.

This allows the user to see that some regions are well studied while other regions contain very little experimental evidence.

8. Suggesting the Next Experiment

Once the data-coverage system is working, FlameLens can identify regions where the model is uncertain and NASA has little experimental data.

The system can then suggest a possible next experimental condition, for example:

Fuel: n-Heptane
Oxygen: 17%
Pressure: 1 atm
Droplet diameter: 2.8 mm
Reason: This region is poorly represented in the current dataset and is close to the predicted extinction boundary.

This should be presented as a data-driven experimental suggestion, not as a claim that NASA should perform that exact experiment.

9. What to Do When the Dataset Is Not Enough

The solution is different for each of the three modules.

Flame Extinction – Build the First ML Model

Use PSI-69 FLEX as the first and main ML dataset.

Recommended inputs include fuel, ambient pressure, ambient gas composition and initial droplet diameter, using only fields that are actually available and meaningful for the selected dataset.

Recommended first target: Extinction vs Non-Extinction, rather than inventing a three-class 'Sustained / Near Extinction / Extinction' label if the source data does not directly provide those labels.

Recommended model comparison:

Logistic Regression – simple baseline

Random Forest – main first nonlinear model

Gradient Boosting – optional comparison

Gaussian Process – later experiment for uncertainty-aware prediction if the implementation is practical

Random Forest is a good starting point because the dataset is small, tabular and nonlinear. A neural network is not justified by the current dataset size.

Flame Spread – Do Not Force ML Yet

The current BASS-II and SAFFIRE tables are not sufficient for a reliable general flame-spread ML model. The current data should first be expanded and a clean target such as spread rate should be obtained from the appropriate analyzed NASA data.

The next practical step is to investigate additional BASS/BASS-II analyzed data and the relevant NASA PSI-62 material so that the final training table can contain consistent inputs and measured spread-rate outputs.

Until enough labeled data is available, the Flame Spread page can operate as a NASA Experiment Explorer:

User enters or selects conditions.

System finds similar BASS/BASS-II/SAFFIRE experiments.

System displays the actual experimental observations.

No unsupported ML prediction is displayed.

Once a sufficiently large and consistent spread-rate dataset is available, the same ML pipeline can be introduced.

Soot/Smoke – First Establish the Correct Target

The current SPICE dataset is more promising in size, but the current experimental table does not provide a simple target such as 'Soot = Yes/No'. Therefore, the first step is to obtain or identify the appropriate analyzed SPICE measurement that can be used as a target.

Possible target types depend on what the analyzed NASA data actually contains, such as a smoke-point-related measurement, flame length or another experimentally measured soot-related quantity.

Only after the target is clearly defined should a regression or classification model be trained.

10. Recommended Overall ML Architecture

The project should not be designed as three mandatory ML models. Instead, it should use a task-specific prediction layer.

NASA PSI Data
↓
Data Normalization
↓
Task-Specific Dataset
↓
Check Data Coverage
↓
If sufficient → ML Prediction
If insufficient → NASA Evidence / Experiment Search
↓
Similarity + Confidence + Feature Importance
↓
Physics + RAG Explanation
↓
FlameLens Website

This architecture allows the project to remain useful even when one dataset is too small for ML.

11. Practical Development Plan

Complete the FLEX extinction dataset cleaning and target definition.

Train Logistic Regression and Random Forest as the first comparison.

Evaluate using cross-validation and report the actual performance instead of assuming the model is good.

Add SHAP or feature importance to explain the prediction.

Implement nearest NASA experiment retrieval.

Implement the data-coverage check before prediction.

Add what-if analysis for changing experimental variables.

Investigate BASS/BASS-II analyzed data and PSI-62 for a proper flame-spread target.

Investigate analyzed SPICE data and define a scientifically valid soot/smoke target.

Only then expand the ML models to Flame Spread and Soot/Smoke.

12. Final Position of the Project

FlameLens should not claim that it can predict every type of microgravity flame behaviour. The stronger and more realistic claim is that it is an AI + Physics research platform that uses NASA combustion experiments to predict experimentally supported behaviour, retrieve similar experiments, identify data gaps and explain the result.

The strongest version of the project is therefore not 'we trained Random Forest on NASA data'. It is: 'FlameLens combines machine learning with NASA experimental evidence and physics-aware interpretation, while explicitly identifying when the available data is insufficient for a reliable prediction.'